

CENG322 - Video Assignment 1

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1. What are the advantages and disadvantages of open-source software development based on the examples given in the documentary?

- Advantages:
 - Collaboration: Open source software allows all developers to collaborate which leads to improving the product developed. For example, Linux was developed by hundreds of programmers which made it a successful operating system.
 - Cost-Effectiveness: Open source software is available for free, which reduces costs. This is touched upon in the documentary by comparing Linux to expensive systems like Windows.
 - Security: Since the source code is open, anyone can examine it, which helps to quickly fix security vulnerabilities. Even today, with updates, a security vulnerability that may be found in Linux can be fixed immediately because the code is open to everyone.
 - Flexibility: Users have the freedom to change and distribute the software. Today's Linux distros can be an example of this. We can make customizations that we cannot make in other operating systems with different distros.
- Disadvantages:
 - Complexity: Open source software can be complicated to install and use, especially for new users. The documentary mentions that Linux requires expertise to initially install. For example, when I first switched to Linux, it took me a long time to learn.
 - Monetization: While open source software can be freely distributed, it can be difficult to monetize. Companies like Red Hat have managed to do this by offering support and services, but this may not work for everyone.
 - Official Support: While there is community support, it may not always be as reliable or timely as professional support. This can be a hassle for users.

2. What part of the documentary was the most interesting for you? Why?

The part of the documentary that generally interests me is the story of Linus Torvalds. Because it shows how collaboration can lead to incredible achievements, even when starting from something small and personal. It is very surprising that he embarked on such an adventure at my age in Finland, and that an operating system he was going to develop for his personal use grew to compete with Microsoft. While doing this, he shared it as open source and invited others to contribute. It is amazing that people from all over the world came together and designed something that is still developing. Although I have been using Linux for about a year, learning the history of this operating system that I use has added a lot to me. An important lesson I learned from the video was that many people came together on a project and developed it openly without competing.

3. Can you identify areas in your own academic or personal projects where adopting open-source principles might be beneficial? Explain your reasoning.

Using open source principles in my own academic or personal projects; It can be useful for adding new features, building community, and fixing bugs. For example, we have an project this semester in CENG318 course. By making this project open source, we can collaborate with other developers and add more new features. It can also be useful for fixing bugs in the project. If this project continues in real life (becomes an application), the openness of the codes will also create trust among users. The absence of any data theft etc. can help us gain the trust of users.